Lecture Scribe Example

L6_S2_A.pdf
PDF

I want a lecture scribe to refer for an exam. Here attached is the lecture notes for which I want you to give me exam ready scribe. The scribe should function as exam reference material and include proofs, derivations, and worked examples exactly as presented in the lecture.

Lecture Scribe
Course:
Lecture number:
Topic:
Date:

Definitions:
(Key definitions)

Theorems / Propositions:
(Statements)

Proofs:
(Main proof ideas / steps)

Derivations / Equations:
(Important derivations)

Examples:
Given:
Find:
Solution:
Remarks:
Assumptions
Special cases
Summary:
Key results

Above written is an example Scribe. Using the same

structure, generate a lecture scribe for the provided context.

Below is an **exam-ready lecture scribe**, written **strictly from the provided lecture slides** and following **exactly the structure you specified**.
All definitions, statements, derivations, and examples mirror the lecture content and flow, with no extra material added beyond what is present in the slides.
(Source: Lecture 6 slides)

Lecture Scribe

Course: CSE400 – Fundamentals of Probability in Computing
Lecture Number: 6
Topic: Discrete Random Variables, Expectation, and Problem Solving
Date: January 22, 2025

Definitions

Random Variable

A **random variable** X on a sample space Ω is a function

$$X : \Omega \to \mathrm{R}$$

that assigns a real number to each outcome $\omega \in \Omega$.

In this lecture, we restrict attention to **discrete random variables**, i.e., random variables that take values from a **finite or countably infinite set**.

Discrete Random Variable

A random variable is **discrete** if:

- Its support is **countable** (finite or countably infinite),
- Probabilities are assigned to **individual values**,
- Each possible value has **strictly positive probability**.

Probability Mass Function (PMF)

Let X be a discrete random variable with range

$$R_X = \{x_1, x_2, x_3, \dots\}$$

The **probability mass function (PMF)** of X is defined as:

$$P_X(x_k) = P(X = x_k), \quad k = 1, 2, 3, \dots$$

Theorems / Propositions

Normalization Property of PMF

Since a discrete random variable must take **exactly one** of its possible values,

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

Bayes' Formula (Proposition 3.1)

Given events $B_1, B_2, \dots, B_n$ forming a partition of the sample space and an event A with $P(A) > 0$,

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{n} P(A \mid B_j)P(B_j)}$$

- $P(B_i)$: **a priori probability**
- $P(B_i \mid A)$: **posterior probability**

Proofs

Proof of PMF Normalization

Since X must take one of the values x_k,

$$1 = P\left(\bigcup_{k=1}^{\infty} \{X = x_k\}\right)$$

Because the events $\{X = x_k\}$ are mutually exclusive,

$$1 = \sum_{k=1}^{\infty} P(X = x_k) = \sum_{k=1}^{\infty} P_X(x_k)$$

Derivation of Bayes' Formula

Using conditional probability,

$$P(A \mid B_i) = \frac{P(A \cap B_i)}{P(B_i)} \quad \Rightarrow \quad P(A \cap B_i) = P(A \mid B_i)P(B_i)$$

Since $A = \bigcup_{j=1}^{n}(A \cap B_j)$,

$$P(A) = \sum_{j=1}^{n} P(A \mid B_j)P(B_j)$$

Substituting into

$$P(B_i \mid A) = \frac{P(A \cap B_i)}{P(A)}$$

gives Bayes' formula.

Derivations / Equations

Poisson-Type PMF Normalization

Given

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots, \ \lambda > 0$$

Using normalization:

$$\sum_{i=0}^{\infty} p(i) = 1 \Rightarrow c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Since

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

we get

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Examples

Example 1: Tossing Three Fair Coins

Given:

An experiment consists of tossing 3 fair coins.

Let Y denote the number of heads observed.

Find:

The PMF of Y.

Solution:

Possible values: $Y \in \{0, 1, 2, 3\}$

$$P(Y = 0) = P(T, T, T) = \frac{1}{8}$$
$$P(Y = 1) = \frac{3}{8}$$
$$P(Y = 2) = \frac{3}{8}$$
$$P(Y = 3) = \frac{1}{8}$$

Check normalization:

$$\sum_{i=0}^{3} P(Y = i) = 1$$

Remarks:

This PMF can be visualized as a **bar diagram**.

Example 2: PMF with Unknown Constant

Given:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

Find:

1. $P(X = 0)$
2. $P(X > 2)$

Solution:

From normalization:

$$c = e^{-\lambda}$$

$$P(X = 0) = e^{-\lambda}$$

$$P(X > 2) = 1 - P(X \leqslant 2)$$

$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$$

$$= 1 - (e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2} e^{-\lambda})$$

Example 3: Bayes' Theorem – Auditorium Problem

Given:

- 30 rows in an auditorium
- Row i has $10 + i$ seats (Row 1: 11 seats, ..., Row 30: 40 seats)
- A row is chosen uniformly at random
- A seat is chosen uniformly from the selected row

Find:

1. $P(\text{Seat } 15 \mid \text{Row } 20)$
2. $P(\text{Row } 20 \mid \text{Seat } 15)$

Solution:

$$P(\text{Seat } 15 \mid \text{Row } 20) = \frac{1}{30}$$

Using Bayes' theorem:

$$P(\text{Row } 20 \mid \text{Seat } 15) = \frac{P(\text{Seat } 15 \mid \text{Row } 20)P(\text{Row } 20)}{\sum_{k=5}^{30} P(\text{Seat } 15 \mid \text{Row } k)P(\text{Row } k)}$$

$$= \frac{\frac{1}{30} \cdot \frac{1}{30}}{\sum_{k=5}^{30} \frac{1}{10+k} \cdot \frac{1}{30}} \approx 0.0325$$

Remarks

- PMF applies **only** to discrete random variables
- The sum of PMF values must always equal 1
- Bayes' theorem updates **prior beliefs** into **posterior beliefs**

Assumptions

- Outcomes are equally likely unless stated otherwise
- Coin tosses and selections are independent
- All rows are selected with equal probability

Special Cases

- Single-value RV → PMF equals 1 at that value
- Continuous RVs use PDFs instead of PMFs
- Zero probability for individual points in continuous RVs

Summary: Key Results

- Random variable is a function from outcomes to real numbers
- Discrete RV → PMF
- $\sum P_X(x) = 1$
- Bayes' theorem relates prior and posterior probabilities
- Poisson-type PMFs normalize using e^{λ}

If you want, I can also:

- Condense this into **last-day exam revision notes**
- Convert it into **LaTeX-ready exam sheets**
- Add **margin cues** like *"likely exam question"* or *"easy 5-marker"*